Academic Year 2025/2026

Adaptive Interactive Systems

Master Project

Design and Implementation of an Adaptive Interactive System

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Project Objective

The objective of this project is to design and implement an **adaptive interactive system** in a domain of your choice. The system should dynamically adjust its **recommendations** and **interface presentation** based on the **user profile** and relevant **contextual information**.

You will progressively develop your system using a **provided base structure** as a reference framework. Your task is to adapt this framework to your selected domain and incrementally integrate additional functionalities and adaptation logic, following the guidelines described in this document.

The provided base project, **available for download on Moodle, includes:**

- An SQLite database containing data on users and rules
- A basic adaptation engine
- A starter frontend with templates and static assets for adaptive content display

Final submission:

You are required to submit the complete **source code** of your system along with a **project report** detailing your design choices, implementation process, and evaluation results. The projects will be presented during the last class session.

Possible application domains include (but are not limited to):

- **E-commerce & Consumer Experience**
  - Adaptive Shopping Assistant: Recommends products depending on user preferences, browsing history, and contextual data (e.g., time of day, device, location).
  - Personalized Promotion Engine: Adjusts discounts and marketing messages based on customer loyalty and engagement.
- **Health, Lifestyle & Well-being**
  - Adaptive Fitness Coach: Suggests workouts and adjusts intensity based on physical profile, previous activity, and real-time context (e.g., weather, sleep).
  - Personalized Nutrition Advisor: Recommends meal plans according to user goals (weight loss, muscle gain), allergies, and time constraints.
- **Knowledge, Productivity & Professional Tools**
  - Context-Aware Task Manager: Reorganizes to-do lists or notifications according to user workload, schedule, and level of focus.
  - Adaptive Career Recommendation System: Suggests professional opportunities or skills to develop based on background, aspirations, and job market data.
- **Mobility, Environment & Smart Cities**
  - Adaptive Travel Planner: Recommends routes or transport modes depending on weather, traffic, user preferences (eco-friendly, fastest), and schedule.
  - Smart Energy Dashboard: Adjusts visualizations and advice according to user profile (resident, business) and energy consumption habits.

Team composition: The project must be completed in **groups of two students**.

Step 1 – Design Adaptive System

Objective: Conceptualize your adaptive system before starting development.

Instructions:

- Explore and run the provided source code to simulate the functioning of the adaptive quiz system.
- Study the project structure and understand how users, rules, and the frontend interact.
- Identify the domain for which you want to generate recommendations.
- Sketch initial interface ideas to guide your development.

Questions to consider:

- What is the target user population in your domain?
- Which types of recommendations or tasks will your system adapt?

Step 2 – Enriched User Profile

Objective: Extend the user model to include static and dynamic attributes capturing user behavior or interactions with the system.

Instructions:

1. Start from the existing `users`, `preferences` and `performance` tables in `users.db`.
2. Identify additional static or dynamic attributes relevant to your domain, such as:
    - Time spent per task or per session
    - Frequency of engagement
    - Success rate or other performance metrics
3. Implement automatic updates of these attributes after each session.
4. Consider how these attributes influence adaptation. Examples inspired by the quiz system:
    - Low persistence → assign shorter but more frequent tasks
    - Fast completion → increase difficulty or provide bonus content
5. Use SQL commands to read/write dynamic data in the user model.

Questions to consider:

- Which user characteristics and behaviors are most relevant for adaptation in your domain?
- How will you structure the database to efficiently store and update this information?

Step 3 – Context Model

Objective: Integrate **contextual data** into the adaptive system.

Instructions:

1. Create a `Context` table in SQLite with fields such as:
    - Device type (desktop, tablet, mobile)
    - Available time
    - Connection quality or other environmental factors
2. Implement functionality to:
    - Insert or update context data dynamically for each user
    - Retrieve the current context when generating recommendations

Questions to consider:

- Which contextual factors influence user experience in your domain?
- How will you keep context data updated during a session?

Step 4 – Adaptation Rules

Objective: Implement adaptation logic based on user profile and context.

Instructions:

1. Start with simple if/else rules to adapt recommendations to user profile and context.
2. Store rules in the `Rules` table in SQLite.
3. Examples inspired by the quiz system:
    - Beginner + short time → easy tasks with few items
    - Advanced + high performance → challenging tasks
4. Develop a parser to evaluate conditions and apply adaptations automatically.
5. Test the system with two users in different contexts, producing different outputs. Ensure your system combine user profile and context to determine adaptations.

Questions to consider:

- How will your parser evaluate conditions efficiently at runtime?
- How will you validate that adaptation rules produce meaningful differences?

Step 5 – Adaptive Interface

Objective: Visualize adaptations in the interface.

Instructions:

1. Use the provided frontend templates and static files as a starting point.
2. Implement dynamic interface changes according to user profile and context. Examples inspired by the quiz system:
    - Desktop → multiple items per screen

    - Mobile → one item per screen
    - Slow connection → reduce multimedia content
3. Compare user experience across different scenarios using the adaptation rules.

Questions to consider:

- Which user and context attributes are most relevant for interface adaptation?
- Which UI elements are the most important to adapt in your domain?

Step 6 – History and Dashboard (Optional / Advanced)

Objective: Track user progress and refine adaptation using historical data.

Instructions:

1. Create a `LearningHistory` table to store session data. Examples inspired by the quiz system:
    - Score, attempts, time spent, date
2. Visualize progress using tables or charts in the frontend.
3. Optionally, use historical data to influence adaptation rules. Examples:
    - High average score → increase difficulty
    - Low success rate → provide additional guidance

Final Deliverables

At the end of the project, submit:

1. **Source code** including backend, database, and frontend
2. **Report** detailing:
    - How the structure was adapted to your domain
    - Implementation of user and context models
    - Application of adaptation rules
    - Interface design and adaptive behavior
    - Optional: dashboard and analytics

You should demonstrate that:

- The same user receives different experiences depending on context.
- The system relies on user profile, context, and adaptation rules.
- Adaptations are visible in the interface.
- The interface itself adapts dynamically to the user and context.